Naren Prakash

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Katy TX, US

Research Interests

Time series forecasting and anomaly detection; Bayesian and sequential inference; topic modeling and NLP over scientific corpora; reproducible and accessible statistical computing.

Education

B.S. Statistics and Data Science Los Angeles CA, US
University of California, Los Angeles June 2025
GPA: 3.833 / 4.00

Coursework: Probability Theory, Mathematical Statistics, Linear Models, Regression Analysis, Experimental Design, Monte Carlo Methods, Statistical Computing and Optimization, Applied Geostatistics, Text Mining, Statistical Consulting.

Research Experience

Statistical Consultant (Capstone Practicum) Los Angeles CA, US
UCLA Statistical Consulting Center March 2025 - June 2025

- Scoped institutional-factors question for UCLA Academic Planning and Budget into an analysis plan; sole owner of that question within a five-person team; presented recommendations to client.
- Assembled data layer from College Scorecard and NCES IPEDS records (2004–2023) joined on CIP codes; cost-adjusted earnings against scraped county living-wage estimates; imputed missing fields by chained equations with random forests.
- Derived peer groups by HDBSCAN clustering; ranked index drivers by cross-validated LASSO and random-forest importance.
- Delivered findings as an interactive Dash/Plotly dashboard with pinned dependencies, reproducible by client unaided.

Undergraduate Researcher Los Angeles CA, US
Department of Statistics, UCLA (advisor: Vivian Lew) January 2025 - June 2025

- Designed a study of methodological change across 136,238 arXiv records; fixed estimation approach and analysis plan before running anything; deduplicated records and measured how dropping incomplete records shifted corpus composition.
- Built a hybrid topic-classification pipeline over LDA, BERTopic, and transformer embeddings; required agreement across all three methods before treating an assignment as an estimation input.
- Trained and validated a LightGBM model forecasting research-output trends by topic; evaluated stability across topic groups of very different sizes.

Researcher and Analyst Los Angeles CA, US
Bruin Sports Analytics, UCLA September 2024 - June 2025

- Estimated NFL route-target probabilities by Bayesian updating over nflfastR play-by-play data, posteriors revised sequentially as observations accumulated.
- Designed recurring match-report visualizations of rate and frequency estimates; adopted by UCLA men's tennis coaching staff for opponent scouting.

Professional Experience

Statistics Curriculum Consultant (supervisor: Vivian Lew) Remote
Department of Statistics, UCLA June 2026 - Present

- Own end-to-end design of a Bayesian statistics module for an upper-division UCLA course; translate current research literature into lecture material, worked estimation examples, and graded assignments packaged for instructor reuse.
- Write original expositions of each method's estimation approach and assumptions under which it holds, with original diagrams of estimator and posterior behavior, sufficient for an instructor to teach the method from the write-up alone.
- Devised per-student parameterized assignments — variant parameters hashed from student identifier, answers checked against known analytic results — resistant to answer sharing and language-model submissions.
- Built authoring toolchain solo (see Software) and a test-backed Jupyter notebook assignment deployed in Canvas.

Data Management and Strategy Houston TX, US
LivaNova May 2025 - April 2026

- Established recurring measurement of enterprise cloud consumption: selected telemetry that actually measured cost and utilization, chose which statistics to publish, and fixed how each estimate would be computed before building.
- Trained time series forecasting and anomaly-detection models on consumption data to project future values and flag observations inconsistent with expected distributions; surfaced idle and over-provisioned spend.
- Built ELT pipelines in Azure Synapse Analytics and Azure Data Factory consolidating multiple subscriptions, replacing a monthly email report and cutting reporting lag from roughly 30 days to near real time.
- Produced published estimates in SQL, Python, and Power BI and the dashboards reporting them, adopted across finance, IT, and leadership; documented data lineage, model assumptions, and limitations for non-technical reviewers.

Data Science and Machine Learning R&D Intern Houston TX, US
Halliburton June 2024 - September 2024

- Researched time series modeling for predictive maintenance over six to nine months of second-resolution truck-sensor telemetry, targeting failures that put field operators at risk of injury.
- Built and trained a transformer model for sequential anomaly detection from scratch, reaching 92% precision at 95% accuracy on held-out data, under a validation split chosen to keep training and evaluation periods from overlapping in time.
- Benchmarked competing estimators for the same forecasting problem by reproducing published approaches against established baselines on common held-out data; documented which performed best under which conditions.
- Wrote a Dask multiprocessing pipeline generating and streaming synthetic training data matched to observed telemetry distribution, clearing a throughput bottleneck on CUDA-accelerated training.

Data Analyst Intern Houston TX, US
Neoage Services June 2023 - September 2023

- Automated bank-statement generation with a Python pipeline, cutting production time by roughly 95% relative to prior manual process.
- Wrote a parser extracting structured fields from unstructured raw statement pages, reducing manual transcription errors.

Presentations

1. **Naren Prakash**. Research on Research on Research: Analyzing Historical Trends in Statistical and Computational Research. *Undergraduate Research Week, UCLA*, Los Angeles CA, US; Contributed talk, presenter.
2. **Naren Prakash**. A Bayesian Approach to Play Prediction. *Bruin Sports Analytics Showcase, UCLA*, Los Angeles CA, US; Contributed talk, presenter.

3. **Naren Prakash**. Autoencoders for Predictive Maintenance. *Halliburton technical seminar*, Houston TX, US; Internal seminar talk, presenter.

Software

figure-gate
Python; sole author — June 2026 - Present

- CI-gated, pip-packaged pipeline generating statistical diagrams — distributions, posterior updates, uncertainty intervals — under colorblind-accessible palettes (Okabe-Ito, Viridis).
- Written as authoring toolchain for the UCLA Bayesian statistics module (see Professional Experience); packaged and licensed for reuse outside the course.

Interactive statistics teaching site
Astro, TypeScript, Rust — naren-p.com — June 2026 - Present

- Built distribution explorers in D3, with live editable Python running in-browser through a Pyodide web worker.
- Wrote Rust corpus-preparation tool behind an in-browser grader scoring free-text explanations against on-device WASM embeddings.
- Gated every push behind three CI stages: type checks, lint, Vitest units, Playwright and axe-core suites, supply-chain auditing.

Teaching and Mentoring

Mentor — Los Angeles CA, US
Statistics and Data Science Club, UCLA — September 2024 - June 2025

- Advised undergraduate students on course planning, research opportunities, and career preparation in statistics and data science.

Module Director — Los Angeles CA, US
CityLab at UCLA — March 2022 - March 2024

- Directed hands-on biotechnology lab sessions for visiting high school students; co-designed instructional modules taught in them.

Skills

Statistics: Regression and generalized linear models | Bayesian and sequential inference | Experimental and study design | Monte Carlo and resampling methods | Missing-data methods
Modeling: Time series analysis and forecasting | Anomaly and outlier detection | Gradient-boosted trees | Topic modeling (LDA, BERTopic) | Geostatistics (variograms, kriging)
Programming: Python (NumPy, Pandas, scikit-learn, PyTorch, Transformers, Dask) | R (base, Tidyverse) | SQL | Rust | TypeScript
Data systems: Microsoft Azure (Synapse Analytics, Data Factory) | ETL/ELT pipeline design
Graphics: matplotlib | D3 | Plotly/Dash | Power BI | Colorblind-safe and accessible statistical graphics
Tools: Git | CI/CD | Linux/Unix | CUDA | LaTeX and Typst | Agentic coding with Claude Code
Spoken languages: English (native) | Tamil (receptive) | Spanish (basic) | Korean (basic)

References

Vivian Lew, Senior Continuing Lecturer, Department of Statistics and Data Science, UCLA
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Nicolas Christou, Senior Continuing Lecturer, Department of Statistics and Data Science, UCLA
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